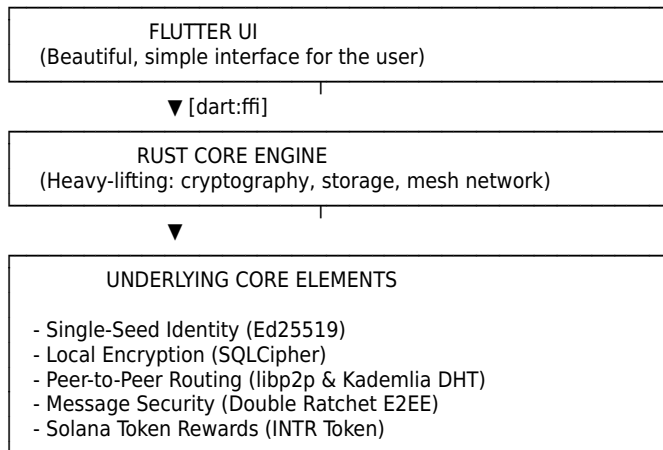


To understand **Introvert** from the ground up, think of it as a complete ecosystem that eliminates the middleman. Traditional messaging apps rely on massive corporate servers to verify who you are, store your messages, and track your data.

Introvert replaces that entire structure using a few highly specialized tools. Here is a simple breakdown of the core elements being used and exactly why we use them.

🔧 The Core Technical Stack



1. 🦀 The Rust Core Engine & Dart FFI

- **What it is:** Rust is a high-performance, ultra-secure programming language that handles our backend. `dart:ffi` is the bridge that links our frontend directly to Rust.
- **Why we use it:** We want our security, networking, and encryption to run at native speed without any lag or memory leaks. Rust does the heavy lifting, while Flutter creates the interface.

2. 🔑 Single-Seed Identity (Ed25519)

- **What it is:** A single 12-word seed phrase that generates your master cryptographic keys.
- **Why we use it:** Instead of using passwords, phone numbers, or email addresses, your identity is pure math. This single seed generates your chat identity, unlocks your local database, and acts as your on-chain wallet.

3. 🛡️ SQLCipher (Encrypted Local Storage)

- **What it is:** A highly secure, encrypted version of a standard local database.
- **Why we use it:** Since there is no central server, your chat history is stored directly on your device. SQLCipher ensures that even if someone steals your physical device, they cannot read your database without your master key.

4. 🌐 libp2p & Kademlia DHT (The Mesh Network)

- **What it is:** libp2p is a framework for peer-to-peer networking, and Kademlia DHT is like a decentralized phone book.
- **Why we use it:** Instead of sending your messages through a corporate server (like WhatsApp or Telegram), your device connects directly to other peers. The Kademlia DHT helps your app find exactly where your friends are on the internet without revealing their real-world IP addresses.

5. 🔒 The Double Ratchet (End-to-End Encryption)

- **What it is:** A dynamic encryption system that changes the security keys after *every single message* sent or received.

- **Why we use it:** It guarantees complete security. If a hacker somehow manages to steal one of your temporary message keys, they still cannot read your past messages (Forward Secrecy) or decrypt any future messages (Break-in Recovery).

6. The Introvert Token (INTR)

- **What it is:** A custom digital currency minted directly on the Solana Mainnet.
- **Why we use it:** Because there are no corporate servers to pass messages along, we rely on independent "Anchor Nodes" (run by desktop users) to act as digital mailboxes. The **INTR** token acts as a rewards system—automatically paying those Anchor nodes for their storage and network uptime.

INTRT the Introvert Token. A custom SPL token minted on Solana. Used strictly as the payment currency for the platform. Users pay it to Anchor Nodes to reward them for their message caching, storage, and server uptime.